13. Older Adults: Standards of Care in Diabetes—2026

*American Diabetes Association Professional Practice Committee for Diabetes**

Diabetes Care 2026;49(Suppl. 1):S277–S296 | https://doi.org/10.2337/dc26-S013

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Recommendations

13.1 Assess the medical, psychological, functional (self-management abilities), and social domains in older adults with diabetes using a comprehensive approach to determine goals and therapeutic approaches for diabetes management. **B**

13.2 Screen at least annually for geriatric syndromes (e.g., cognitive impairment, depression, urinary incontinence, falls, persistent pain, and frailty), hypoglycemia, and polypharmacy in older adults with diabetes, as they may affect diabetes management and diminish quality of life. **B**

Diabetes is a highly prevalent health condition in the aging population. Over 29% of people over the age of 65 years have diabetes (1,2). The number of older adults living with diabetes is expected to increase rapidly in the coming decades as the population ages and the effects of increasing obesity rates manifest in this population (3). Importantly, diabetes in older adults is a highly heterogeneous condition. While type 2 diabetes predominates in the older population (as in the younger population), improvements in insulin delivery, technology, and care over the last few decades have led to increasing numbers of people with childhood and adult-onset type 1 diabetes surviving and thriving into their later decades. Type 1 diabetes is increasingly diagnosed in older adulthood as well, with greater awareness of and attention to appropriate diabetes classification across the life course, with higher incidence in Western countries (4).

Diabetes management in older adults requires regular assessment of medical, psychological, functional, and social domains. When assessing older adults with diabetes, it is important to accurately categorize the type of diabetes, diabetes duration, the presence of complications and comorbidities, the individual's capacity for self-management and availability of support systems, treatment burden, and treatment-related concerns, such as fear of hypoglycemia, polypharmacy, and financial barriers. Screening for

**A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at https://doi.org/10.2337/dc26-SINT.*

This section has received endorsement from the American Geriatrics Society.

Duality of interest information for each contributor is available at https://doi.org/10.2337/dc26-SDIS.

Suggested citation: American Diabetes Association Professional Practice Committee for Diabetes. 13. Older adults: Standards of Care in Diabetes—2026. Diabetes Care 2026;49(Suppl. 1):S277–S296

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diabetes complications in older adults should be individualized and periodically revisited, as this may affect treatment goals and therapeutic approaches (5–7). Older adults with diabetes have higher rates of functional disability, accelerated muscle loss, mobility impairment, frailty, and coexisting illnesses, such as hypertension, chronic kidney disease, coronary heart disease, stroke, and premature death than those without diabetes (2,8). They also have higher rates of common geriatric syndromes such as cognitive impairment, depression, urinary incontinence, falls, persistent pain, frailty, and polypharmacy (1). These conditions may affect older adults' diabetes self-management abilities and quality of life, particularly if unaddressed, and older adults with diabetes often require greater caregiver support than those without diabetes (2,9–11). See section 4, "Comprehensive Medical Evaluation and Assessment of Comorbidities," for the full range of issues to consider when caring for older adults with diabetes.

The Institute for Healthcare Improvement has developed an evidence-based "4Ms" framework for age-friendly health care that is being adopted by many health systems caring for older adults. The key elements of this approach are Mentation, Medications, Mobility, and What Matters Most, with each element affecting the others (12). The 4Ms framework provides a helpful conceptual model to address person-specific issues that may be interrelated and affect diabetes management in older individuals (**Fig. 13.1**) and offers a framework to establish individualized treatment goals and approaches (13–15), including delivery of diabetes self-management education (particularly when complicating factors arise or when transitions in care occur) and whether the current plan is too complex for the individual's self-management ability or for the care partners providing care (16). Particular attention should be paid to complications that can develop over short periods of time and/or would significantly impair functional status, such as visual and lower-extremity complications. **Table 13.1** suggests a structured approach to screening for common geriatric syndromes and other causes of functional impairment using key screening questions, followed by validated screening tools when specific geriatric syndromes are suspected, as well as resources for scoring. These tools should be used when clinically indicated and can be performed by any member of the diabetes care team. Importantly, the general care principles discussed here apply to all individuals with complex health needs and are not limited to older adults.

NEUROCOGNITIVE FUNCTION

Recommendation

13.3 Screening for early detection of mild cognitive impairment or dementia should be performed for adults 65 years of age or older at the initial visit, annually, and as appropriate. **B**

Older adults with diabetes are at higher risk of cognitive decline and institutionalization than older adults without diabetes (17–19). Presentation of cognitive impairment ranges from subtle executive dysfunction to memory loss to overt dementia. People with diabetes have higher incidences of all-cause dementia, Alzheimer disease, and vascular dementia than people without diabetes (20). People with Alzheimer disease or other dementias are more likely to develop diabetes than those without it; this possible link emphasizes critical metabolic processes that contribute to neurodegeneration (21). Both hyperglycemia and hypoglycemia are associated with a decline in cognitive function (22,23), and longer duration of diabetes is associated with worsening cognitive function. A newly

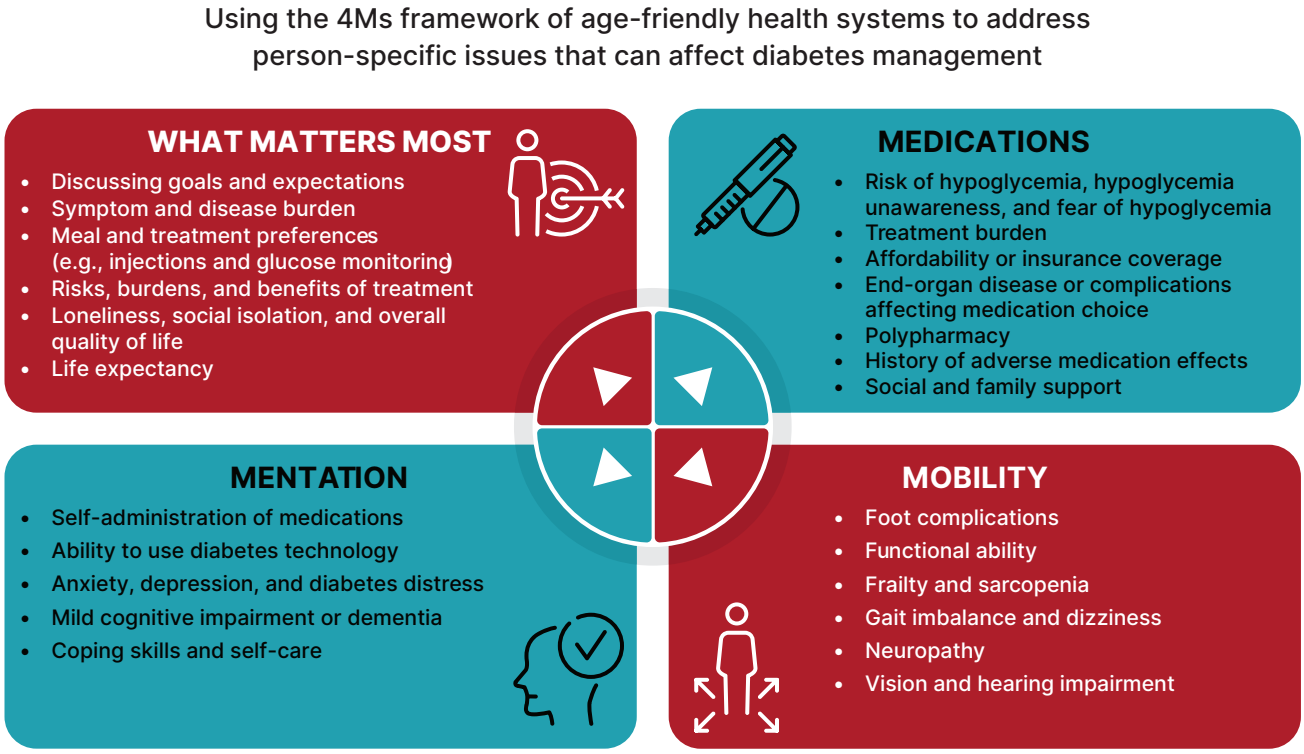

Figure 13.1—Using the 4Ms framework of age-friendly health systems to address person-specific issues that can affect diabetes management.

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Table 13.1—Geriatric syndromes and other functional impairments: key symptoms and suggested screening approaches

Domain	Sample screening questions	Suggested screening measure	No. of items	Score range	Cut points	Time to complete	Measure translated languages
Geriatric syndromes							
Cognitive impairment or dementia	Have you noticed any changes in your memory or ability to think clearly?	Mini-Cog (31)	5	0–5	<3 validated for dementia screening	3 min	Multiple
Delirium	What day of the week is it?	4AT (182)	4	0–12	≥4 possible delirium	2 min	Multiple
Depression	During the past month, have you often been bothered by feeling down, depressed, or hopeless?	Geriatric Depression Scale 5 (GDS-5) (183,184)	5	0–5	≥2 possible depression	2 min	Multiple
Falls and risk of falling	Have you fallen in the past year or are you afraid of falling due to a balance or walking problem?	STEADI tool (185)	3	0–3	A single "yes" response to any question indicates an increased risk of falling	1–2 min	Multiple
Frailty	Have you felt unusually tired recently or had difficulty walking up steps without resting?	Clinical Frailty Scale (CFS) (186)	1	1–9	≥5 clinically meaningful frailty	1–2 min	Multiple
Malnutrition	Have you unintentionally lost weight in the last 6 months?	Mini Nutritional Assessment Short Form (MNA-SF) (187)	6	0–14	12–14, normal nutritional status; 8–11, at risk for malnutrition; 0–7, malnourished	5 min	Multiple
Pain	Are you currently experiencing any pain?	Numeric Pain Rating Scale	1	0–10	1–3, mild; 4–6, moderate; 7–10, severe	1–2 min	NA
Polypharmacy	Are you taking more medications than you think you should, or are you experiencing side effects?	Beers Criteria or review all prescription and nonprescription medications (188)	List review	NA	The presence of any medication on the list warrants review, especially if the risk outweighs the benefit	Varies	Multiple
Sarcopenia	Have you had difficulty walking across a room or climbing a flight of stairs in the past year?	SARC-F Questionnaire (189)	5	0–10	≥4 sarcopenia risk	2–3 min	Multiple
Sensory impairment: hearing	Do you have difficulty hearing, even when using a hearing aid?	Whisper test (190)	6	NA	Possible hearing loss: unable to repeat at least 3 of the random letters and numbers	2–3 min	NA

Continued on p. S280

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Table 13.1—Continued

Domain	Sample screening questions	Suggested screening measure	No. of items	Score range	Cut points	Time to complete	Measure translated languages
Sensory impairment: vision	Do you have difficulty seeing, even when wearing your glasses?	Snellen chart	1	NA	20/40 or worse indicates vision impairment	2–3 min	NA
Sleep disorders: insomnia	Do you have trouble falling asleep, staying asleep, or waking up too early and not being able to fall back asleep?	Insomnia Severity Index 7 (ISI-7) (191)	7	0–28	≥8 insomnia	3 min	Multiple
Urinary incontinence	Do you have trouble with bladder control or accidental leakage?	3 Incontinence Questions (3IQ) (192)	3	NA	The tool classifies type of incontinence based on responses	1–2 min	Multiple
Functional impairments							
Dexterity	Do you have trouble using your hands for everyday tasks like buttoning clothes?	Button and coin test: button and unbutton a shirt button and pick up a small coin from a flat surface	2	0–2	0, significant impairment; 1, mild to moderate impairment; 2, limited to no impairment	1–2 min	NA
Dizziness	Do you experience dizziness, feeling lightheaded, or a feeling of spinning (vertigo)? Have you noticed any changes in your balance or steadiness?	Orthostatic vital signs; neurological examination (e.g., assessment of gait, balance, reflexes, sensory loss, vestibular function); medical evaluation	NA	NA	NA	NA	NA
Executive functioning	Do you find it hard to plan or organize your daily activities?	Trail Making Test B	25	Time to finish	≥180 s indicates impairment in executive function	3–5 min	Multiple

This table summarizes key symptoms and suggested screening approaches for common geriatric syndromes and other functional impairments that can be present in older adults with diabetes. NA, not applicable.

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recognized clinical entity, diabetes-related dementia, is emerging as distinct from Alzheimer disease and vascular dementia. Diabetes-related dementia is characterized by a slower progression of dementia, absence of typical neuroimaging findings, advanced age, elevated A1C levels, long duration of diabetes, high frequency of insulin use, frailty, sarcopenia, and dynapenia (loss of muscle strength not caused by neurologic or muscular diseases) (23). Ongoing studies are evaluating whether lifestyle interventions may help to maintain cognitive function in older adults (24). However, studies on diabetes prevention or intensive glycemic and blood pressure management have not demonstrated a reduction in cognitive decline (25,26). Several glucose-lowering drugs, such as thiazolidinediones, glucagon-like peptide 1 receptor agonists (GLP-1 RAs) and sodium–glucose cotransporter 2 (SGLT2) inhibitors, have shown small benefits on slowing progression of cognitive decline (27). A systematic review and meta-analysis found that although cardioprotective glucose-lowering therapies were not associated with a reduction in all-cause dementia, GLP-1 RAs were associated with a statistically significant reduction in all-cause dementia (28). Notably, cardiovascular risk factors are also associated with an increased risk of cognitive decline and dementia. Management of blood pressure and cholesterol lowering with statins have been associated with a reduced risk of incident dementia and are, thus, particularly important in older adults with diabetes.

Older adults with diabetes should be carefully screened and monitored for cognitive impairment (2). Several assessment tools are available to screen for cognitive impairment (29,30), such as the Mini-Cog (31), Mini-Mental State Examination (32), and the Montreal Cognitive Assessment (33), which may help identify individuals with dementia (who are experiencing memory loss, a decrease in executive function, and declines in their basic and instrumental activities of daily living) or those requiring neuropsychological evaluation when dementia is suspected. Annual screening is indicated for adults 65 years of age or older for early detection of mild cognitive impairment or dementia (6,34). Screening for cognitive impairment should also be performed when an individual presents with a significant decline in clinical status due to increased problems with self-care activities and medication management, such as errors in calculating insulin dose, difficulty counting carbohydrates, skipped meals, skipped insulin doses, and difficulty recognizing, preventing, or treating hypoglycemia. People who screen positive for cognitive impairment should receive diagnostic assessment as appropriate, including referral to a behavioral health professional for formal cognitive and neuropsychological evaluation if indicated and feasible (35).

Identifying cognitive impairment early has important implications for diabetes care. The presence of cognitive impairment can make it challenging to reach individualized glycemic, blood pressure, and lipid goals. Cognitive dysfunction may make it difficult for individuals to perform complex self-care tasks (29), such as monitoring glucose, administering and adjusting insulin doses, and maintaining the timing and nutritional content of meals. These factors increase risk for hypoglycemia, which, in turn, can worsen cognitive function and have multiple other adverse effects in older individuals with diabetes. When cognitive dysfunction is identified, it is therefore essential to simplify care plans and engage the appropriate support structure to assist aging individuals with diabetes in all aspects of their care.

HYPOGLYCEMIA

Recommendations

13.4 Ascertain and address episodes of hypoglycemia at routine visits because older adults with diabetes have a greater risk of hypoglycemia, especially when treated with hypoglycemic agents (e.g., sulfonylureas, meglitinides, and insulin). **B**

13.5 Recommend continuous glucose monitoring (CGM) for older adults with type 1 diabetes **A** and type 2 diabetes on insulin therapy **B** to improve glycemic outcomes, reduce hypoglycemia, and reduce treatment burden.

13.6 Consider the use of automated insulin delivery systems **B** and other advanced insulin delivery devices such as connected pens **E** to reduce risk of hypoglycemia for older adults, based on individual ability and support system.

Older adults may be at higher risk of hypoglycemia for many reasons, including irregular meal intake, particularly in the setting of requiring insulin therapy, and worsening kidney function (36). As described above, older adults have higher rates of cognitive impairment and dementia, leading to difficulties in performing complex self-care activities (e.g., glucose monitoring, insulin administration and dose adjustment). Cognitive decline has been associated with increased risk of hypoglycemia, and conversely, severe hypoglycemia has been linked to increased risk of dementia (37–39). Therefore, as discussed in Recommendation 13.3, it is important to routinely screen older adults for cognitive impairment and dementia and discuss findings with the individuals and their care partners.

People with diabetes and their care partners should be routinely queried about any history of hypoglycemic events, impaired hypoglycemia awareness, and fear of hypoglycemia as discussed in section 6, "Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises." Symptoms and signs of hypoglycemia are also discussed in section 6 and are of value to care partners of individuals with cognitive impairment who may not be able to report hypoglycemia symptoms. Older adults can also be stratified for future risk for hypoglycemia with validated risk calculators (e.g., Kaiser Hypoglycemia Model for adults with type 2 diabetes) (40) and with consideration of hypoglycemia risk factors (**Table 6.5**). An important step to mitigate hypoglycemia risk is to determine whether the person with diabetes is skipping meals or has difficulty correctly taking and dosing their glucose-lowering medications. Glycemic goals and pharmacologic treatments may need to be adjusted to minimize the occurrence of hypoglycemic events, prioritizing use of medications at low risk for hypoglycemia and other adverse effects (41). This recommendation is also supported by results from multiple randomized controlled trials (RCTs), such as the Action to Control Cardiovascular Risk in Diabetes (ACCORD) study and the Veterans Affairs Diabetes Trial (VADT), which showed that intensive treatment protocols aimed to achieve an A1C <6.0% with complex drug plans significantly increased the risk for hypoglycemia requiring assistance compared with standard treatment (42,43). However, these intensive treatment plans included extensive use of insulin and minimal use of GLP-1 RAs, and they preceded the availability of SGLT2 inhibitors. Hypoglycemia

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may also be precipitated by acute illness and other stressful events such as trauma or surgery. During these events, older adults and their care partners should be provided individualized guidance on glycemic monitoring and adjustment of glucose-lowering medications to prevent hypoglycemia. See INTERCURRENT ILLNESS in section 6, "Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises," for more details.

Use of Continuous Glucose Monitoring and Advanced Insulin Delivery Devices

Continuous glucose monitoring (CGM) devices have been shown to be effective at improving glycemic management and acceptable to people of all age-groups, including older individuals with type 1 diabetes (44–48) or insulin-requiring type 2 diabetes (49,50). Additionally, benefits have been shown with CGM use in older individuals not treated with insulin, particularly for reducing hypoglycemia and improving other glycemic outcomes. Importantly, it may take longer for older adults to learn and gain facility with new technology and it may be beneficial to engage their caregivers in the process. Moreover, it is important to assess a person's cognitive and functional capacity for using technology and to ensure availability of engagement and support of the care partner, if appropriate (51). CGM may also be useful for glucose monitoring in older adults with physical and/or cognitive limitations who require monitoring of blood glucose by a surrogate and in those who reside in group homes, assisted living facilities, or other post-acute long-term care settings.

Advanced insulin delivery devices have similarly been shown to improve glycemic outcomes in small studies conducted among older adults. The Older Adult Closed Loop (ORACL) trial in 30 older adults (mean age 67 years) with type 1 diabetes found that an automated insulin delivery (AID) strategy was associated with significant improvements in time in range (TIR), as well as modest improvements in hypoglycemia, compared with sensor-augmented pump therapy (52). In a different open-label, crossover design clinical trial in 37 older adults (aged ≥60 years), 16 weeks of treatment with a hybrid closed-loop advanced insulin delivery system improved TIR and reduced time above range compared with sensor-augmented pump therapy; however, in contrast to the ORACL study, no significant differences in hypoglycemia were observed (53). Both studies enrolled older individuals whose blood glucose was relatively well managed at baseline (mean A1C ~7.4%). A later RCT of older adults with type 2 diabetes using multiple daily injections who were unable to manage insulin therapy on their own demonstrated an increase of TIR of 27% over 12 weeks of AID use in addition to tailored home health care services (54). More recently, in the Automated Insulin Delivery in Elderly With Type 1 Diabetes (AIDE T1D) study that included 82 adults aged 65 years and older, participants randomized to AID use experienced less hypoglycemia than those randomized to hybrid closed loop, predictive low glucose suspend, and sensor-augmented pump therapy in a crossover design (55). In the extension phase of the study, 91% of the participants chose the AID system (55). Finally, a real-world evidence analysis of a Medicare population (4,243 individuals, 89% with type 1 diabetes, mean age 67.4 years) also indicated that initiating hybrid closed-loop insulin delivery was associated with improvements in mean glucose and a 10% increase in TIR (56). These studies provide evidence that older individuals with type 1 and type 2 diabetes can successfully use advanced insulin delivery technologies to improve glycemic outcomes, as has been seen in younger populations.

TREATMENT GOALS

Recommendations

13.7a Older adults with diabetes with few and stable coexisting chronic illnesses, and intact cognitive and functional status, should have lower glycemic goals (such as A1C <7.0–7.5% [<53–58 mmol/mol]) and/or time in range [TIR] 70–180 mg/dL [3.9–10.0 mmol/L] of ≥70% and time below range ≤70 mg/dL [≤3.9 mmol/L] of ≤4%) if CGM is used. **C**

13.7b Older adults with diabetes and intermediate or complex health are clinically heterogeneous with variable life expectancy. Selection of glycemic goals should be individualized and should prioritize avoidance of hypoglycemia, with less stringent goals (such as A1C <8.0% [<64 mmol/mol] and/or TIR 70–180 mg/dL [3.9–10.0 mmol/L] of ≥50% and time below range <70 mg/dL [3.9 mmol/L] of <1%) for those with significant cognitive and/or functional limitations, frailty, severe comorbidities, and a less favorable risk-to-benefit ratio of diabetes medications. **C**

13.7c Older adults with very complex or poor health receive minimal benefit from stringent glycemic goals. Clinicians should focus on avoiding hypoglycemia and symptomatic hyperglycemia. **C**

13.8 Screening for diabetes complications should be individualized in older adults with diabetes. Prioritize screening for complications that would lead to impairment of functional status or quality of life. **C**

13.9 The on-treatment blood pressure goal for most older adults with diabetes is <130/80 mmHg when it can be achieved safely, **A** and more a relaxed blood pressure goal (e.g., <140/90 mmHg) may be used for people with poor health, limited life expectancy, or high risk for adverse effects of hypertensive therapy. **E**

13.10 Treatment of other cardiovascular risk factors should be individualized in older adults with diabetes, considering the time frame of benefit. Lipid-lowering therapy and antiplatelet agents may benefit those with life expectancies at least equal to the time frame of primary prevention or secondary intervention trials. **A**

As for all people with diabetes, diabetes self-management education and ongoing diabetes self-management support are vital components of diabetes care for older adults and their caregivers. Self-management knowledge and skills should be reassessed following a significant clinical change or hospitalization, when treatment plan changes are made, or when an individual's functional abilities diminish. In addition, declining or impaired ability to perform diabetes self-care behaviors may be an indication that an older person with diabetes needs a referral for cognitive and physical functional assessment, using age-normalized evaluation tools, as well as help establishing a support structure for diabetes care (5,35). See **Table 13.1** for screening and evaluation tools for geriatric syndromes and other functional impairments.

The care of older adults with diabetes is complicated by their clinical, cognitive, and functional heterogeneity and their

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varied prior experience with disease management. Some older individuals may have developed diabetes years earlier and have significant complications, others are newly diagnosed and may have had years of undiagnosed diabetes with resultant complications, and still, other older adults may have truly recent-onset disease with few or no complications (57). Some older adults with diabetes have other underlying chronic conditions, substantial diabetes-related comorbidity, limited cognitive or physical functioning, or frailty (58,59). Other older individuals with diabetes have little comorbidity and intact cognitive and physical functioning.

Life expectancy is affected by the age of the individual, disease burden, and degrees of frailty and disability. Multiple prognostic tools for life expectancy for older adults are available (60,61). The Life Expectancy Estimator for Older Adults with Diabetes (LEAD) tool was developed and validated among older adults with diabetes, and a high risk score was strongly associated with having a life expectancy of <5 years (62). These data may be a useful starting point to inform decisions about selecting less stringent glycemic goals, particularly when more stringent goals may necessitate use of pharmacotherapies with higher risk of hypoglycemia and other adverse effect profiles (62,63). Older adults also vary in their preferences for the intensity and mode of glucose management (64). Health care professionals caring for older adults with diabetes must take this heterogeneity into consideration when engaging people with diabetes in shared decision-making to establish treatment goals (14,15,41) (**Table 13.2**). In addition, older adults with diabetes should be assessed for disease treatment and self-management knowledge, health literacy, and mathematical literacy (numeracy) at the onset and throughout treatment. Limited time for medical visits, and competing priorities such as acute problems or change in living situation or social support, can make implementation of these recommendations challenging. See **Fig. 6.1** for individual/disease-related factors to consider when determining individualized glycemic goals.

A1C results may be inaccurate in those who have received blood transfusions and who have medical conditions that affect red blood cell turnover (see section 2, "Diagnosis and Classification of Diabetes," for additional details on the limitations of A1C) (65). Conditions affecting red blood cell turnover that are common in older adults include kidney failure, recent significant blood loss, and erythropoietin therapy. In these instances, blood glucose monitoring and/or CGM should be used for glycemic goal setting (**Table 13.2**). Serum glycated protein assays such as fructosamine may also be useful for glycemic monitoring in conjunction with other measures (see section 6, "Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises") (66–68).

Older Adults With Good Functional Status and Without Complications

There are few long-term studies in older adults demonstrating the benefits of intensive glycemic, blood pressure, and lipid management. Older adults who can be expected to live long enough to realize the benefits of long-term intensive diabetes management, who have good cognitive and physical function, and who experience little (or acceptable to them) burden of treatment may be treated using therapeutic interventions and goals similar to those for younger adults with diabetes (**Table 13.2**).

Older Adults With Complications and Reduced Functionality

Older adults with diabetes categorized as having complex or intermediate health (**Table 13.2**) are still heterogeneous with respect to their function and life expectancy (69–71). Based on concepts of competing mortality and time to benefit, individuals with shorter life expectancy, advanced diabetes complications, life-limiting comorbid illnesses, frailty, or substantial cognitive or functional impairments will have less benefit from glucose lowering and should have less stringent glycemic goals (72). These individuals may not only not derive benefit from intensive glucose lowering but are also more likely to experience serious adverse effects of treatment, such as hypoglycemia (73). However, significant hyperglycemia should also be avoided—even in the setting of complex health status—as elevated glucose levels (>180 mg/dL [>10 mmol/L]) increase risks of dehydration, weakness, infection, poor wound healing, and hyperglycemic crises (74). Glycemic goals should, at a minimum, avoid these consequences. Factors to consider for individualizing glycemic goals are outlined in **Fig. 6.1** and **Fig. 13.1**. Clinicians should also consider the balance of risks and benefits of an individual's diabetes medications, including disease-specific benefits (such as reducing symptomatic heart failure or stabilizing chronic kidney disease) and burdens such as hypoglycemia risk, tolerability, difficulties of administration, inadequate support system, and financial cost. In addition, attention to oral health, vision and hearing loss, foot care, fall prevention, and early detection of depression will improve quality of life.

While **Table 13.2** provides overall guidance for identifying individuals with complex and very complex health needs, there is no global consensus on the optimal classification strategy. Empiric research on the classification of older adults with diabetes based on comorbid illness has suggested three broad classes of individuals: a healthy, a geriatric, and a cardiovascular class (14,69,75). The geriatric class has the highest prevalence of obesity, hypertension, arthritis, and incontinence, while the cardiovascular class has the highest prevalence of myocardial infarctions, heart failure, and stroke. Compared with the healthy class, the cardiovascular class has the highest risk of frailty and subsequent mortality. Additional research is needed to develop a reproducible classification scheme to distinguish the natural history of disease as well as differential response to glucose management and specific glucose-lowering agents (76).

Older Adults at the End of Life

For people with diabetes receiving palliative care and end-of-life care, the focus should be to avoid hypoglycemia and symptomatic hyperglycemia while reducing the burdens of glycemic management. Thus, as organ failure develops, the treatment plan should be deintensified. At the end of life, most agents for type 2 diabetes may be removed (77). There is, however, no consensus for the end-of-life management of type 1 diabetes (78). Consultation with a geriatric or palliative care specialist might be warranted to assist with complex medical and functional issues as well as advance care planning. See END-OF-LIFE CARE, below, for additional information.

Beyond Glycemic Management

Although minimizing hyperglycemia may be important in older individuals with

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Table 13.2—Framework for considering treatment goals for glycemia, blood pressure, and dyslipidemia in older adults with diabetes

Characteristics and health status of person with diabetes	Rationale	Reasonable A1C goal*	Reasonable CGM goals	Fasting or preprandial glucose	Bedtime glucose	Blood pressure	Lipids
Healthy (few coexisting chronic illnesses, intact cognitive and functional status)	Longer remaining life expectancy	<7.0–7.5% (<53–58 mmol/mol)	TIR 70–180 mg/dL (3.9–10.0 mmol) of ≥70%, and TBR ≤70 mg/dL (≤3.9 mmol/L) of ≤4%	80–130 mg/dL (4.4–7.2 mmol/L)	80–180 mg/dL (4.4–10.0 mmol/L)	<130/80 mmHg	Statin, unless contraindicated or not tolerated
Complex/intermediate (multiple coexisting chronic illnesses† or two or more ADL impairments or mild to moderate cognitive impairment)	Variable life expectancy. Individualize goals, considering: • Severity of comorbidities • Cognitive and functional limitations • Frailty • Risk-to-benefit ratio of diabetes medications • Individual preference	<8.0% (<64 mmol/mol)	TIR 70–180 mg/dL (3.9–10.0 mmol) of ≥50% and TBR <70 mg/dL (<3.9 mmol/L) of <1%	90–150 mg/dL (5.0–8.3 mmol/L)	100–180 mg/dL (5.6–10.0 mmol/L)	<130/80 mmHg	Statin, unless contraindicated or not tolerated
Very complex/poor health (PALTC or end-stage chronic illnesses‡ or moderate to severe cognitive impairment or two or more ADL impairments)	Limited remaining life expectancy makes benefit minimal	Avoid reliance on A1C; glucose management decisions should be based on avoiding hypoglycemia and symptomatic hyperglycemia		100–180 mg/dL (5.6–10.0 mmol/L)	110–200 mg/dL (6.1–11.1 mmol/L)	<140/90 mmHg	Consider likelihood of benefit with statin

This table represents a consensus framework for considering treatment goals for glycemia, blood pressure, and dyslipidemia in older adults with diabetes. The characteristic categories are general concepts. Not every individual will clearly fall into a particular category. Consideration of individual and care partner preferences, care partner engagement, abilities, and resources is an important aspect of treatment individualization. Additionally, an individual's health status and preferences may change over time. ADL, activities of daily living; CGM, continuous glucose monitoring; PALTC, post-acute and long-term care; TBR, time below range; TIR, time in range. *A lower A1C goal may be set for an individual if achievable without recurrent or severe hypoglycemia or undue treatment burden. †Coexisting chronic illnesses are conditions serious enough to require medications or lifestyle management and may include arthritis, cancer, heart failure, depression, emphysema, falls, hypertension, incontinence, stage 3 or worse chronic kidney disease, myocardial infarction, and stroke. "Multiple" means at least three, but many individuals may have five or more (193). ‡The presence of a single end-stage chronic illness, such as stage 3–4 heart failure or oxygen-dependent lung disease, chronic kidney disease requiring dialysis, or uncontrolled metastatic cancer, may cause significant symptoms or impairment of functional status and significantly reduce life expectancy. Adapted from Kirkman et al. (5).

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diabetes, greater reductions in morbidity and mortality are likely to result from a clinical focus on comprehensive cardiovascular risk factor modification. There is strong evidence from clinical trials of the value of treating hypertension in older adults (79–81), with treatment of hypertension to individualized blood pressure goals indicated in most. There is less evidence for lipid-lowering therapy and aspirin therapy, although the benefits of these interventions for primary and secondary prevention are likely to apply to older adults whose life expectancies equal or exceed the time frames of the clinical trials (82). In the case of statins, the follow-up time of clinical trials ranged from 2 to 6 years. While the time frame of trials can be used to inform treatment decisions, a more specific concept is the time to benefit for a therapy, which, for statins, is 2.5 years (83).

LIFESTYLE MANAGEMENT

Recommendations

13.11a Recommend healthful eating with adequate protein intake (at least 0.8 g/kg body weight/day) for older adults with diabetes to maintain and potentially higher, individualized amounts to regain lean body mass and function. **B**

13.11b Recommend regular physical activity, including aerobic activity, weight-bearing exercise, and resistance training as tolerated in those who can safely engage in such activities, particularly to maintain lean body mass, especially in those pursuing weight loss. **B**

13.12 For older adults with type 2 diabetes, overweight or obesity, and the capacity to exercise safely, an intensive lifestyle intervention focused on dietary changes, physical activity, and weight loss (e.g., 5–7%) should be considered for its benefits on quality of life, mobility and physical functioning, and cardiometabolic risk. **A**

Lifestyle management in older adults should be tailored to frailty status. Diabetes in the aging population is associated with reduced muscle strength, poor muscle quality, and accelerated loss of muscle mass, which may result in sarcopenia or dynapenia (84) and/or osteopenia (85). Diabetes is also recognized as an independent risk factor for frailty. Frailty is characterized by a decline in physical performance and an increased risk of negative health outcomes due to physiological vulnerability and functional or psychosocial stressors. Inadequate nutritional intake, particularly inadequate protein intake, can increase the risk of sarcopenia and frailty in older adults. There is no current evidence showing that adjusting protein intake beyond the general recommendations (0.8–1.5 g/kg body weight/day or 15–20% of calories) improves glycemic management or cardiovascular outcomes in individuals with diabetes (86,87). Nevertheless, individualized protein goals may be beneficial, as some research suggests that moderately higher protein intake (20–30%) can support diabetes management by enhancing satiety (88). In populations with sarcopenia, interventions combining protein supplementation with resistance exercise have resulted in increases in muscle mass and muscle strength (89). Special attention should be paid to malnutrition or the risk of malnutrition in older adults with diabetes given its association with sarcopenia (90,91). Malnutrition is also associated with decreases in activities of daily living, grip strength, physical performance of lower limbs, cognition, and quality of life (92–94). See section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes," for a description of malnutrition and screening recommendations. Management of malnutrition, sarcopenia, and frailty in diabetes includes optimal nutrition with adequate protein intake combined with a physical activity program that includes aerobic, weight-bearing, and resistance training. The benefits of a structured physical activity program (as in the Lifestyle Interventions and Independence for Elders [LIFE] study) in frail older adults include reducing sedentary time, preventing mobility disability, and reducing frailty (95). The goal of these programs is not weight loss but enhanced functional status. For nonfrail older adults with type 2 diabetes and overweight or obesity, an intensive lifestyle intervention designed to reduce weight is beneficial across multiple outcomes. The Look AHEAD (Action for Health in Diabetes) trial is described in section 8, "Obesity and Weight Management for the Prevention and Treatment of Diabetes." Look AHEAD specifically excluded individuals with a low functional status. It enrolled people between 45 and 74 years of age and required that they be able to perform a maximal exercise test (96,97). Although the Look AHEAD trial did not achieve its primary outcome of reducing cardiovascular events, the intensive lifestyle intervention produced multiple clinical benefits, including weight loss, improved physical fitness, increased HDL cholesterol, lowered systolic blood pressure, reduced A1C levels, reduced waist circumference, and reduced need for medications (98). Subgroup analyses found improved cardiovascular outcomes among participants who lost at least 10% of baseline body weight in the first year (99). Risk factor management improved with reduced use of antihypertensive medications, statins, and insulin (100). Age-stratified analyses showed that older adults in the trial (aged 60–70 years) experienced benefits similar to those seen for younger participants (101,102). The lifestyle intervention also produced benefits on aging-relevant outcomes, including reductions in multimorbidity, improved physical function, and enhanced quality of life (103–106).

PHARMACOLOGIC THERAPY

Recommendations

13.13 Select medications with low risk of hypoglycemia in older adults with type 2 diabetes, specifically for those with hypoglycemia risk factors. **B**

13.14a Deintensify hypoglycemia-causing medications (e.g., insulin, sulfonylureas, or meglitinides) or switch to a medication class with low hypoglycemia risk for individuals who are at high risk for hypoglycemia, using individualized glycemic goals. **B**

13.14b In older adults with diabetes, deintensify diabetes medications for individuals for whom the harms and/or burdens of treatment may be greater than the benefits, within individualized glycemic goals. **E**

13.14c Simplify complex treatment plans (especially insulin) to reduce the risk of hypoglycemia and polypharmacy and to decrease treatment burden. **B**

13.14d In older adults with type 2 diabetes and established or high risk of atherosclerotic cardiovascular disease, heart failure, and/or chronic kidney disease, the treatment plan should include agents that reduce cardiovascular and kidney disease risk, irrespective of glycemia. **A**

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13.15 Consider costs of care and coverage when developing treatment plans to reduce risk of cost-related barriers to taking medication and performing self-management behaviors. **B**

Special care is required in prescribing and monitoring pharmacologic therapies in older adults, who are at high risk of polypharmacy and its adverse sequelae, often have nuanced considerations for choice of specific medication classes, and may have cognitive and/or functional impairments that affect the capacity for safe self-management. Therapeutic choices should also take into consideration whether older adults with diabetes live independently, have an engaged and available caregiver, or live in a skilled nursing facility, assisted living facility, or group home (and what level of assistance for medication taking and glucose monitoring is available in these facilities). It is important to match complexity of the treatment plan to the self-management ability of older adults with diabetes and their available social and medical support. Cost may be an especially important consideration, as older adults are often treated with multiple medications and live on fixed incomes. Accordingly, the costs of care and insurance coverage rules should be considered when developing treatment plans to reduce the risk of cost-related barriers to use (107,108). See **Fig. 9.4** for general recommendations regarding glucose-lowering treatment for adults with type 2 diabetes, **Table 9.2** for person-and drug-specific factors to consider when selecting glucose-lowering agents, and **Table 9.3** and **Table 9.4** for median monthly cost in the U.S. of noninsulin glucose-lowering agents and insulin, respectively.

Intensive glycemic management in older adults, particularly those with complex health status, with medication plans that increase the risk of hypoglycemia through use of insulin and/or sulfonylureas has been identified as overtreatment, is common in clinical practice (109–113), and may increase the risk of mortality (42). High treatment burden is another indicator of potential overtreatment. Deintensification of glucose-lowering plans can be achieved by either lowering the dose or discontinuing some medications, as long as individualized glycemic goals are maintained (114,115). Multiple studies evaluating deintensification protocols in diabetes and hypertension have shown that deintensification is safe and possibly beneficial for older adults (114). Using the 4S pathway (41), **Fig. 13.2** offers a road map for assessing treatment difficulty; reevaluating glycemic goals through shared decision-making; deintensifying, simplifying, or modifying treatments; and reassessing treatment engagement and burdens.

Metformin

Metformin is a good treatment option for glucose lowering in older adults with type 2 diabetes due to its low risk for hypoglycemia and simple administration profile, but prescription guidelines must be followed carefully and regularly reassessed. Metformin may be used safely in individuals with an estimated glomerular filtration rate (eGFR) $\geq$30 mL/min/1.73 m^2 (116), although lower doses should be used in those with eGFR 30–45 mL/min/1.73 m^2. eGFR should be monitored every 3–6 months in those at risk for decline in kidney function. However, metformin is contraindicated in those with advanced kidney disease and should be used with caution in those with hypoperfusion, hypoxemia, and impaired hepatic function due to potential risk of lactic acidosis. Metformin may be temporarily discontinued before procedures including imaging studies using iodinated contrast, during hospitalizations, and when acute illness may compromise kidney or liver function. Additionally, metformin can cause gastrointestinal side effects (this risk can be minimized through slow uptitration at initiation) and a reduction in appetite that can be problematic for some older adults. Extended-release formulation may be used as an alternative to immediate-release formulation in older adults experiencing difficulties in maintaining medication plans or experiencing gastrointestinal effects. For those taking metformin long term (>4 years), vitamin B12 levels should be monitored annually (117).

Pioglitazone

Pioglitazone has low risk for hypoglycemia but should be used cautiously and in select individuals due to risks for heart failure, fluid retention, weight gain, osteoporosis, and/or macular edema (118,119). Adverse effect risks are reduced with lower doses of pioglitazone and when it is used as part of combination therapy. There may also be the beneficial effect of stroke reduction (120).

Insulin Secretagogues

Sulfonylureas are associated with hypoglycemia, bone loss (121), and fracture risk (122) and should be used with caution. Similarly, meglitinides (repaglinide and nateglinide) increase the risk of hypoglycemia and fracture risk (123,124). If used, sulfonylureas with a shorter duration of action, such as glipizide, may be preferred, although a recent analysis found that the risk of major adverse cardiovascular events was highest with glipizide (125). Glyburide is a longer-acting sulfonylurea with multiple active metabolites and should be avoided in older adults due to its hypoglycemia risk (126). Many antimicrobials (most commonly fluroquinolones and sulfamethoxazole-trimethoprim) interact with sulfonylureas to increase the effective sulfonylurea dose, which may precipitate hypoglycemia (127–129). Sulfonylureas should be reduced or temporarily discontinued in these circumstances. Consistent and adequate eating patterns are essential to reducing hypoglycemia risk, and insulin secretagogues may need to be dose-reduced or held in the setting of illness, fasting, or when eating is otherwise impaired. Older adults treated with insulin secretagogues should be routinely screened for hypoglycemia, hypoglycemia awareness should be assessed, and prevention and treatment strategies should be reinforced.

DPP-4 Inhibitors

Dipeptidyl peptidase 4 (DPP-4) inhibitors have few side effects and minimal risk of hypoglycemia, but their cost may be a barrier to some older adults. DPP-4 inhibitors are relatively weak glucose-lowering agents and do not reduce cardiovascular or kidney disease risks like GLP-1 RAs and SGLT2 inhibitors (130). In general, these medications may be useful in older adults with mild hyperglycemia or with high risk of hypoglycemia when a GLP-1 RA or a dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RA is not tolerated. DPP-4 inhibitors may be used as an alternative to metformin in older adults with low eGFR; linagliptin does not require renal dose adjustment.

GLP-1–Based Therapies

GLP-1 RAs and the GIP/GLP-1 RA tirzepatide are highly effective glucose-lowering

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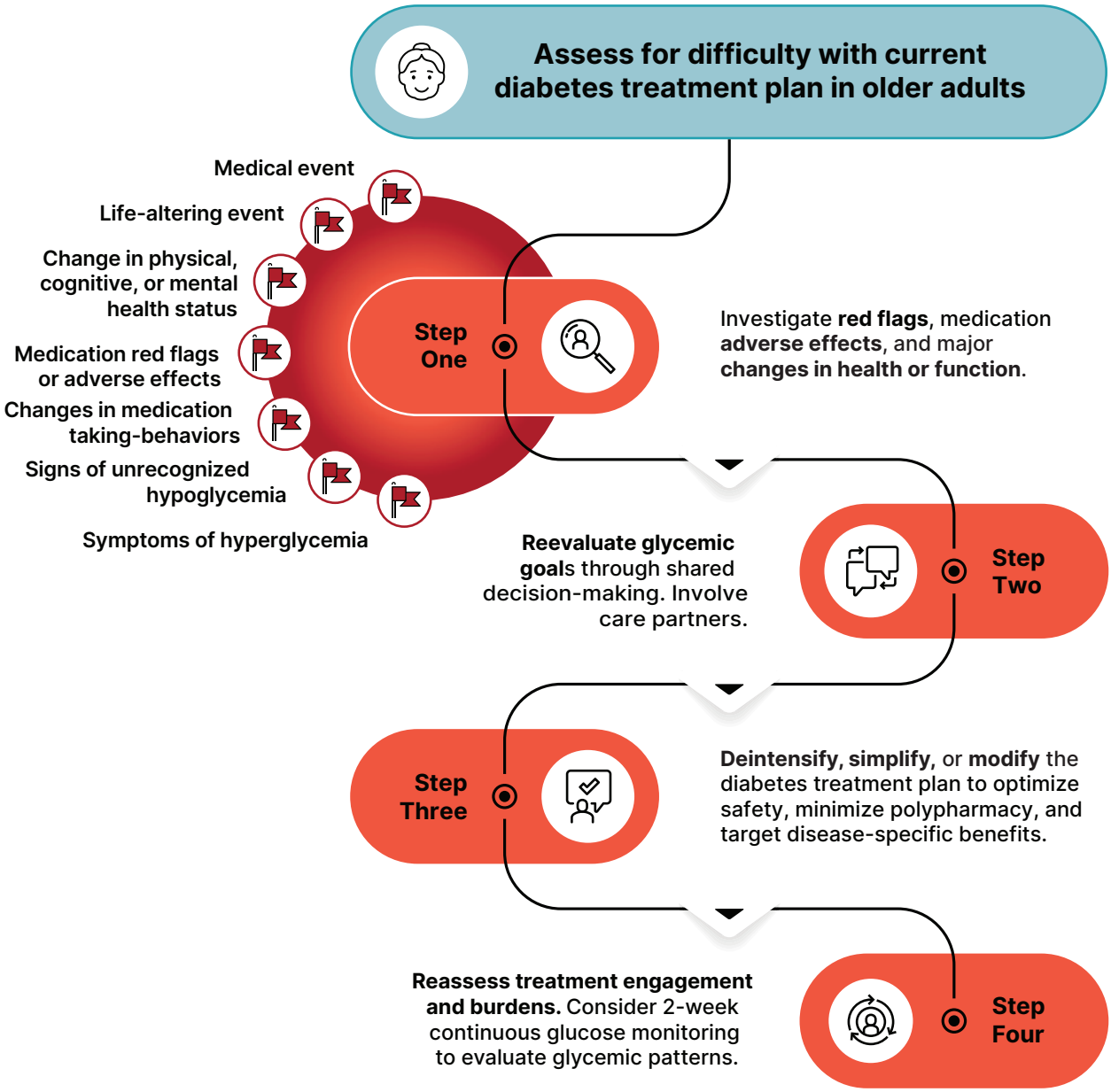

Figure 13.2—Stepwise approach for assessing difficulties in the diabetes treatment plan; reevaluating glycemic goals through shared decision-making; deintensifying, simplifying, or modifying the treatment plan; and reassessing the safety and burdens of any interventions. Created using recommendations from Munshi et al. (41).

medications with low risk for hypoglycemia and can be used in the setting of reduced eGFR, including during dialysis. GLP-1 RAs have demonstrated additional cardiovascular benefits among people with diabetes and established atherosclerotic cardiovascular disease (ASCVD) and those at higher ASCVD risk, with chronic kidney disease, and with symptomatic heart failure with preserved ejection fraction in the setting of obesity. See section 9, "Pharmacologic Approaches to Glycemic Treatment," section 10, "Cardiovascular Disease and Risk Management," and section 11, "Chronic Kidney Disease and Risk Management," for more extensive discussion. In a systematic review and meta-analysis of GLP-1 RA trials, these agents have been found to reduce major adverse cardiovascular events, cardiovascular deaths, stroke, and myocardial infarction to the same degree for people over and under 65 years of age (131). Weight loss achieved with use of GLP-1 RAs and the GIP/GLP-1 RA is also beneficial for obstructive sleep apnea (132), knee osteoarthritis (133), and metabolic dysfunction–associated steatohepatitis (MASH) (134–136). While the evidence for this class of agents for older adults continues to grow, there are a number of practical issues that should be considered specifically for older people. These drugs are injectable agents (with the exception of oral semaglutide) (137), which require visual, motor, and cognitive skills for appropriate administration, although most have a weekly dosing schedule. GLP-1 RAs may also be associated with nausea, vomiting, diarrhea, and/or constipation, and doses should be titrated slowly. GLP-1 RAs are therefore not preferred in older adults experiencing unexplained weight loss or undernutrition or with problematic constipation, significant gastroparesis, recurrent ileus, or bowel obstruction.

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Individuals should be monitored regularly for dehydration and excessive weight loss, which may lead to significant loss of muscle mass (sarcopenia) and bone density. The risk for sarcopenia can be mitigated with adequate protein intake and resistance or strength training (89).

SGLT2 Inhibitors

SGLT2 inhibitors are administered orally, which may be convenient for older adults with diabetes. In those with established ASCVD, these agents have shown cardiovascular benefits (138). This class of agents has also been found to be beneficial for people with heart failure and to slow the progression of chronic kidney disease. See section 9, "Pharmacologic Approaches to Glycemic Treatment," section 10, "Cardiovascular Disease and Risk Management," and section 11, "Chronic Kidney Disease and Risk Management," for a more extensive discussion regarding the indications for this class of agents. Stratified analyses of the trials of this drug class indicate that older adults have similar or greater benefits than younger people (139–141). SGLT2 inhibitors are generally well tolerated among older adults, although thoughtful selection is needed to avoid adverse effects in individuals at elevated risk (142). SGLT2 inhibitors may cause clinically significant volume depletion, for which older adults are at greater risk, and should be used cautiously in older adults who are frail or prone to orthostasis (143). SGLT2 inhibitors cause a higher rate of genital mycotic infections, especially in women, and may need to be discontinued if this effect becomes burdensome (144). Their use may also be associated with a small increase in urinary tract infections; caution should be used in people with recurrent or severe urinary tract infections (144). Because SGLT2 inhibitors typically increase urine volume, symptoms of urinary incontinence should be queried before and after SGLT2 inhibitor initiation (145,146). Euglycemic diabetic ketoacidosis (DKA) is a rare but potentially serious phenomenon associated with treatment with SGLT2 inhibitors, especially in those with multimorbidity who reside in postacute and long-term care (PALTC) settings, with infection being the most common trigger (145,147).

Insulin Therapy

The use of insulin therapy requires that individuals or their caregivers have good visual and motor skills and cognitive ability to manage insulin dosing and administration and to recognize and treat hypoglycemia. Insulin doses, times of administration, and treatment plans should be tailored to meet individualized glycemic goals and to avoid hypoglycemia. Age alone does not limit the utility of technology, and older adults who are capable of using CGM devices or an automated insulin delivery system should be encouraged to use them as long as they help to provide safe and effective glycemic management (see discussion above).

Once-daily basal insulin injection therapy may be a reasonable option in many older adults, although noninsulin therapies are preferred due to lower risk of hypoglycemia and other potential benefits as discussed above. When choosing basal insulin, the prescriber should take into account that long-acting insulin analogs have been associated with a lower risk of emergency department visits and hospitalizations for hypoglycemia than NPH insulin in the Medicare population (148). Multiple daily injections of insulin may be too complex for many older individuals with advanced diabetes complications, life-limiting coexisting chronic illnesses, cognitive or functional impairment, or limited functional status or social support. If affordable, use of insulin pens should be preferred to syringes, mostly in older adults with functional impairment.

Studies suggest that simplification of the insulin plan to match an individual's self-management abilities and their available social and medical support is associated with reduced hypoglycemia and disease-related distress without worsening glycemic outcomes (149–152). **Figure 13.3** depicts an algorithm, and potential approach, that can be used to simplify the insulin administration plan (151), and **Table 13.3** provides examples of and rationale for situations where deintensification and/or insulin plan simplification may be appropriate in older adults.

Other Factors to Consider

The needs of older adults with diabetes and their care partners should be evaluated to construct a tailored care plan. Inadequate social support and reduced access to long-term services and support may reduce these individuals' quality of life and increase the risk of functional dependency (10). The living situation must be considered as it may affect diabetes management and support needs. Social and instrumental support networks (e.g., adult children and care partners) that provide instrumental or emotional support for older adults with diabetes should be included in diabetes management discussions and shared decision-making.

The need for ongoing support of older adults becomes even greater when transitions to PALTC become necessary. Unfortunately, these transitions can lead to discontinuity in goals of care, errors in dosing, and changes in nutrition and activity (153). Older adults in assisted living facilities may not have support to administer their own medications, whereas those living in a nursing home for short-term rehabilitation or PALTC may rely on first-line care partners including nursing and care professionals with variable clinical expertise. Those receiving palliative care (with or without hospice) may require an approach that emphasizes comfort and symptom management while deemphasizing strict metabolic and blood pressure management.

SPECIAL CONSIDERATIONS FOR OLDER ADULTS WITH TYPE 1 DIABETES

Due in part to the success of modern diabetes management, people with type 1 diabetes are living longer, and the population of people over 65 years with type 1 diabetes is growing (154–156). Many of the recommendations in this section regarding comprehensive geriatric assessment and personalization of goals and treatments are directly applicable to older adults with type 1 diabetes; however, this population has unique challenges and requires distinct treatment considerations (157). Insulin is an essential life-preserving therapy for people with type 1 diabetes. To avoid DKA, older adults with type 1 diabetes need some form of basal insulin even when they are unable to ingest meals. Insulin may be delivered through injection, with an AID system, or with an insulin pump alone depending on individual preference, capability, and circumstances. CGM should be used in conjunction with any insulin delivery system. CGM is approved for use by Medicare and can play a critical role in improving A1C, reducing

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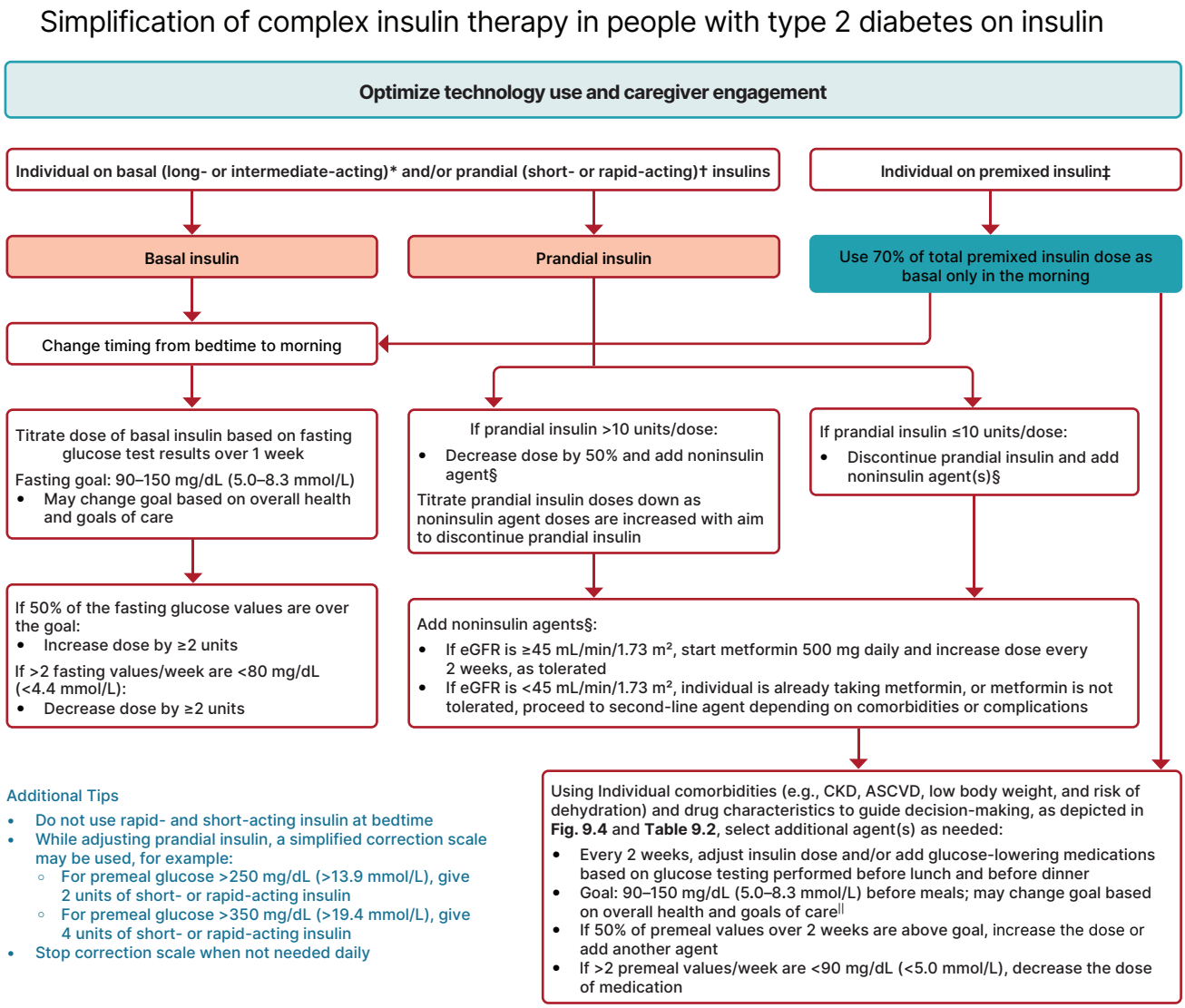

Figure 13.3—Algorithm to simplify insulin administration plans in older individuals. ASCVD, atherosclerotic cardiovascular disease; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate. *Basal insulins: glargine U-100 and U-300, degludec, and human NPH. †Prandial insulins: short-acting (regular human insulin) or rapid-acting (lispro, aspart, and glulisine). ‡Premixed insulins: 70/30, 75/25, and 50/50 products. §Examples of noninsulin agents include metformin, sodium-glucose cotransporter 2 inhibitors, dipeptidyl peptidase 4 inhibitors, glucagon-like peptide 1 receptor agonists (GLP1-RAs), and dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RAs. ||See **Table 13.2**. Adapted with permission from Munshi et al. (151).

glycemic variability, and reducing risk of hypoglycemia (49) (see section 7, "Diabetes Technology," and section 9, "Pharmacologic Approaches to Glycemic Treatment"). In older people with type 1 diabetes, administration of insulin may become more difficult as complications, cognitive impairment, and functional impairment arise. This increases the importance of care partners in the lives of these individuals. Older people with type 1 diabetes may require placement in PALTC settings (i.e., nursing homes and skilled nursing facilities), and unfortunately staff in these settings continue to be less familiar with CGM devices, insulin pumps, or advanced insulin delivery devices. Nevertheless, a feasibility study in PALTC facilities showed that CGM can be useful in older adults with diabetes, although it requires substantial staff training (158). Additional studies of older adults with diabetes living in PALTC facilities using CGM revealed a high prevalence of hypoglycemia both in people using insulin and in those using sulfonylureas, thus showing that this population of older adults in PALTC facilities are at increased risk for hypoglycemia (159,160). Using CGM can provide useful and more prompt information on hypoglycemia as well as time spent in severe hyperglycemia (as noted in a study of 65 residents in 8 long-term facilities) in this population (161). Of note, a recent RCT in PALTC facilities showed that real-time CGM use for up to 60 days was safe and effective in guiding insulin doses compared with BGM by point of care. There were no differences in TIR, time below range, or mean glucose levels (162). Some staff may be less knowledgeable about the differences between type 1 and type 2 diabetes. DKA may be mistaken for sepsis, end-organ failure, or other electrolyte abnormalities. In these instances, the individual or their family may be more familiar with their current diabetes management plan than the staff or health care professionals. Education of relevant support staff and health care professionals in rehabilitation and PALTC settings regarding insulin dosing and use of pumps and CGM

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Table 13.3—Considerations for treatment plan simplification and deintensification/deprescribing in older adults with diabetes

Characteristics and health status of person with diabetes	Reasonable glycemic goal	Rationale/considerations	When may medication plan simplification be required?	When may treatment deintensification be required?
Healthy (few chronic illnesses, intact cognitive and functional status)	A1C <7.0–7.5% (<53–58 mmol/mol)	• Healthy individuals can usually perform complex tasks for glycemic management • During acute illness, individuals may be at risk for medication administration errors	• Severe or recurrent hypoglycemia on insulin regardless of A1C • Wide glucose excursions • Cognitive or functional decline following acute illness	• Severe or recurrent hypoglycemia on insulin or sulfonylureas, regardless of A1C • Wide glucose excursions • Polypharmacy
Complex/intermediate (multiple chronic illnesses or ≥2 ADL impairments or mild to moderate cognitive impairment)	A1C <8.0% (<64 mmol/mol)	• Comorbidities may affect self-management abilities and capacity to avoid hypoglycemia	• Severe or recurrent hypoglycemia on insulin therapy, regardless of A1C • Difficulty managing insulin plan • Significant change in social circumstances, (e.g., loss of care partner or change in living situation or finances)	• Severe or recurrent hypoglycemia on insulin or sulfonylureas, regardless of A1C • Wide glucose excursions • Polypharmacy
Community-dwelling individuals receiving short-term care in a skilled nursing facility	Avoid reliance on A1C; glucose goal 100–200 mg/dL (5.6–11.1 mmol/L)	• Glycemic management is important for recovery, wound healing, and avoidance of infections • Recovery from acute illness may impair cognitive function • More support may be needed on transition to home	• Consider reinstating prehospitalization treatment if it increased in complexity during hospitalization	• Weight loss, anorexia, short-term cognitive decline, and/or loss of physical functioning
Very complex/poor health (PALTC or end-stage chronic illnesses or moderate to severe cognitive impairment or ≥2 ADL impairments)	Avoid reliance on A1C and avoid hypoglycemia and symptomatic hyperglycemia	• No benefits of tight glycemic goals in this population • Hypoglycemia should be avoided • Most important outcomes are maintenance of cognitive and functional status	• The individual would like to decrease the number of injections and finger-stick blood glucose monitoring • The individual has an inconsistent eating pattern	• Cognitive dysfunction, depression, anorexia, or inconsistent eating pattern while taking sulfonylureas or meglitinides • Taking any diabetes medications without clear benefits
At the end of life	Avoid hypoglycemia and symptomatic hyperglycemia	• Goal is to provide comfort and avoid tasks or interventions that cause pain or discomfort • Care partners are important in providing medical care and maintaining quality of life	• Pain or discomfort caused by treatment (e.g., injections or finger sticks) • Excessive stress of care partners due to treatment complexity	• Taking any diabetes medications without clear benefits in improving symptoms and/or comfort

Treatment plan simplification refers to changing strategy to decrease the complexity of a medication plan (e.g., fewer administration times and fewer blood glucose checks) and decreasing the need for calculations (such as sliding-scale insulin calculations or insulin-carbohydrate ratio calculations). Deintensification/deprescribing refers to decreasing the dose or frequency of administration of a treatment or discontinuing a treatment altogether. ADL, activities of daily living; PALTC, post-acute and long-term care. Created using information from Munshi et al. 2016 (151) and 2017 (194).

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is recommended as part of general diabetes education (see Recommendations 13.16 and 13.17).

TREATMENT IN POST-ACUTE AND LONG-TERM CARE SETTINGS

Recommendations

13.16 Recommend diabetes education/training (including that for CGM devices, insulin pumps, and advanced insulin delivery systems) for the staff of post-acute and long-term care settings to improve the management of older adults with diabetes. **E**

13.17 People with diabetes residing in post-acute and long-term care settings need careful assessment of mobility, mentation, medications, and management preferences to establish individualized glycemic goals and to make appropriate choices of glucose-lowering agents and devices (including CGM devices, insulin pumps, and advanced insulin delivery systems) based on their clinical and functional status. **E**

Management of diabetes in the PALTC setting has multiple unique and nuanced considerations. Individualization of health care is important for all people with diabetes; however, practical guidance for feasible and sustainable care delivery is needed for health care professionals, PALTC staff, and care partners (163,164). Training should include diabetes detection, identification of complications, glycemic monitoring, determination of individualized glycemic goals, medication and nutritional considerations, oral care and regular foot assessment, wound and infection concerns, and institutional quality assessment. PALTC facilities should develop their own policies and procedures for prevention, recognition, and management of hypoglycemia and symptomatic hyperglycemia. For more information, see the ADA position statement "Management of Diabetes in Long-term Care and Skilled Nursing Facilities" and diabetes management guidelines for the PALTC setting (153,163,164). With the increased longevity of populations, the care of people with diabetes and its complications in PALTC is an area that warrants greater study.

Nutritional Considerations

An older adult residing in a PALTC facility may have irregular and unpredictable meal consumption, undernutrition, diminished appetite, and impaired swallowing. Furthermore, therapeutic nutrition plans or modified food consistencies may inadvertently lead to decreased food intake and contribute to unintentional weight loss and undernutrition. Meals tailored to a person's culture, preferences, and personal goals may increase quality of life, satisfaction with meals, and nutrition status (165). It may be helpful to give insulin immediately after meals to ensure that the dose is appropriate for the amount of carbohydrate the individual consumed in the meal. However, postprandial insulin administration needs to be balanced against the potential for hypoglycemia, ensuring that prandial insulin is rapid-acting (not short-acting) and is given as soon as possible after eating. It is similarly important that insulin not be given too early before meals and that preprandial glucose monitoring, insulin administration, and eating occur as close to each other as possible.

Hypoglycemia

Older adults with diabetes in PALTC are especially vulnerable to hypoglycemia. They have a disproportionately high number of health conditions that can increase hypoglycemia risk: impaired cognitive and renal function, slowed hormonal regulation and counter-regulation, suboptimal hydration, variable appetite and nutritional intake, requirement for feeding assistance, polypharmacy, and slowed intestinal absorption (166). Oral agents may achieve glycemic outcomes similar to those for basal insulin in PALTC residents with type 2 diabetes (109,167). CGM may be a useful approach to monitoring for and preventing hypoglycemia among individuals treated with insulin in PALTC (160,161,168).

Another consideration for the PALTC setting is that unlike in the hospital setting, health care professionals are not required to evaluate individuals daily. According to federal guidelines, at a minimum, assessments should be done at least every 30 days for the first 90 days after admission and then at least once every 60 days and as clinically indicated for acute medical problems or worsening of existing conditions. Although in practice some individuals may be seen more frequently, individuals may have poorly managed glucose levels or wide excursions without the practitioner being notified. Health care professionals may adjust treatment plans by telephone, fax, or in person directly at the PALTC facilities, provided they are given timely notification of blood glucose management issues from a standardized alert system. It is important for individuals with significant and/or persistent dysglycemia to be reassessed and their treatment modified without delay to prevent recurrence of hypoglycemia and significant hyperglycemia.

The following alert strategy could be considered:

1. Call a health care professional immediately in cases of low blood glucose levels (<70 mg/dL [<3.9 mmol/L]). However, treatment of hypoglycemia should not be delayed.
2. Call as soon as possible when
   a. glucose values are 70–100 mg/dL (3.9–5.6 mmol/L) (treatment plan may need to be adjusted),
   b. two or more blood glucose values >250 mg/dL (>13.9 mmol/L) are observed within a 24-h period accompanied by a significant change in clinical status,
   c. glucose values are consistently >250 mg/dL (>13.9 mmol/L) within a 24-h period,
   d. glucose values are consistently >300 mg/dL (>16.7 mmol/L) over 2 consecutive days,
   e. any reading is too high for the glucose monitoring device, or
   f. the individual is sick, with symptomatic hyperglycemia, vomiting, fever, lethargy, or poor oral intake.

END-OF-LIFE CARE

Recommendations

13.18 When palliative care is needed in older adults with diabetes, health care professionals should discuss goals and intensity of care with people with diabetes and their care partners. Strict glucose, blood pressure, and lipid management are not necessary; consider deintensification or simplification of medication plans and prioritize symptom management. **E**

13.19 Prioritize the overall comfort, prevention of distressing symptoms, and preservation of quality of life and dignity as primary goals for diabetes management at the end of life. **C**

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Management of diabetes in the setting of palliative medicine or hospice care at the end of life is a unique situation. The goal of palliative medicine is to promote comfort, symptom management and prevention (pain, hypoglycemia, hyperglycemia, and dehydration), and preservation of dignity and quality of life in the setting of limited life expectancy (169,170).

In the setting of palliative care, health care professionals should initiate conversations with people with diabetes and their care partners regarding the goals and intensity of diabetes care; strict glucose and blood pressure management may not be consistent with achieving comfort and quality of life. Avoidance of severe hypertension and hyperglycemia aligns with the goals of palliative care. In a multicenter trial, withdrawal of statins among people with diabetes in palliative care was found to improve quality of life (171–173). The evidence for the safety and efficacy of deintensification protocols in older adults is growing for both glucose and blood pressure management (113,174) and is clearly relevant for palliative care. An individual has the right to refuse testing and treatment, whereas health care professionals may consider withdrawing treatment and limiting diagnostic testing, including a reduction in the frequency of blood glucose monitoring (175,176). CGM could be considered when frequent blood glucose monitoring is burdensome but monitoring for hypoglycemia and hyperglycemia is needed. Glycemic goals should aim to prevent hypoglycemia (blood glucose <70 mg/dL) and severe hyperglycemia (blood glucose >250 mg/dL when symptom burden increases), although individuals vary in their response (177). Treatment interventions need to be mindful of quality of life. Careful monitoring of oral intake is warranted. The clinical decision-making process may need to involve the individual, family, and care partners, leading to a care plan that is safe, effective, and goal-concordant (178). Pharmacologic therapy for type 2 diabetes may include oral agents with low likelihood of causing hypoglycemia as first line, followed by a simplified insulin plan such as basal insulin. Agents that can cause gastrointestinal and other adverse symptoms may not be good choices in this setting. As symptoms progress, some agents may be slowly tapered and discontinued.

Different categories have been proposed for diabetes management in those at the end of life (77).

1. A stable individual: Continue with the person's previous medication plan, with a focus on *1*) the prevention of hypoglycemia and *2*) the management of hyperglycemia using blood glucose monitoring, keeping levels below the renal threshold of glucose, and hyperglycemia-mediated dehydration. There is no role for A1C monitoring.
2. An individual with organ failure: Preventing hypoglycemia is of greatest significance. Dehydration must be prevented and treated. In people with type 1 diabetes, insulin administration may be reduced as the oral intake of food decreases but should not be stopped. For those with type 2 diabetes, agents that may cause hypoglycemia should be reduced in dose. The main goal is to avoid hypoglycemia, allowing for glucose values in the upper level of the desired goal range.
3. A dying individual: For people with type 2 diabetes, the discontinuation of all medications may be a reasonable approach, as these individuals are unlikely to have any oral intake. In people with type 1 diabetes, there is no consensus, but a small amount of basal insulin may maintain glucose levels and prevent acute hyperglycemic complications and symptom burden.

Diabetes health care professionals are well positioned to support people with diabetes in advance care planning. Health care professionals can assist people with diabetes in clarifying and documenting their values, preferences, and goals for care in an advanced care plan (172). Advance care plans are guides to help health care professionals and care partners make difficult treatment decisions when the person with diabetes is no longer able to make decisions for themselves. Research shows that people with diabetes want to discuss end-of-life care plans with their health care professional (179). Two validated tools exist to support health care professionals in this process: the Supportive and Palliative Care Indicators Tool (180) and the Gold Standards Framework Proactive Identification Guidance (181).

In conclusion, the management of diabetes in older adults at the end of life necessitates a person-centered approach that prioritizes attainment of individualized treatment goals, comfort, symptom management, quality of life, and the preservation of dignity.

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